

ELECTRICAL POTENTIAL IN SA NODE (PACEMAKER POTENTIAL)

- **Resting Membrane Potential / Pacemaker Potential :—**
 - Pacemaker potential is an **unstable resting membrane potential** (also called **prepotential**).
 - In the SA node, **each impulse automatically triggers the next impulse**.
 - **RMP in SA Node** : -55 to -60 mV (*compared to -85 to -95 mV in ordinary cardiac muscle fibers*).
- **Action Potential Characteristics :—**
 - Depolarization starts slowly until the **threshold level of -40 mV** is reached.
 - Rapid upstroke peaks up to $+5$ mV (**rapid depolarization**), followed by rapid repolarization.

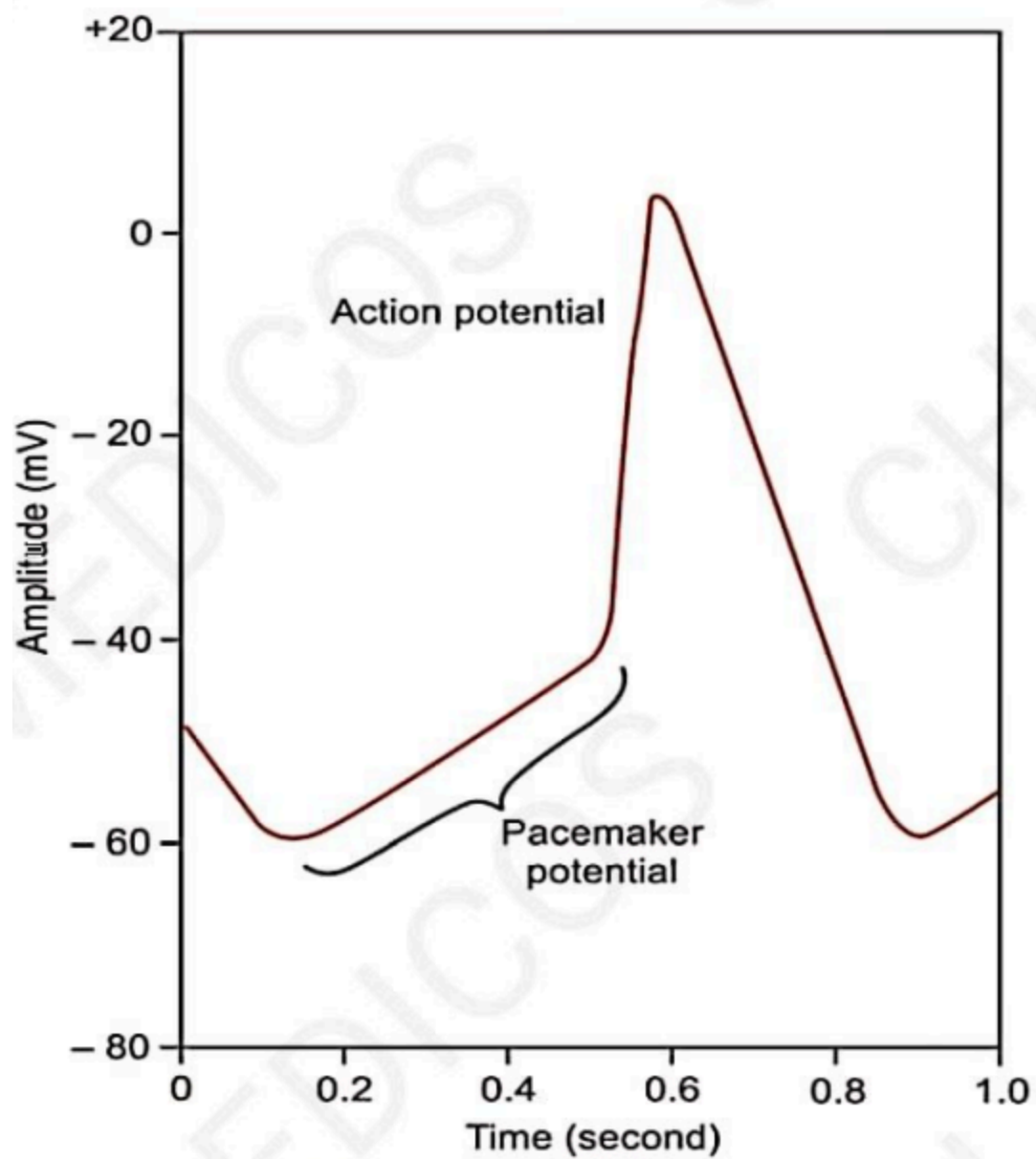


FIGURE 90.3: Pacemaker potential

IONIC BASIS OF ELECTRICAL ACTIVITY IN

PACEMAKER

1. Pacemaker Potential (Prepotential)

Na^+ ions leak into
pacemaker fibers (I_f /
Funny current)

Slow initial depolarization

Transient T-type Ca^{2+}
channels open

Slow Ca^{2+} influx continues
depolarization to
threshold (-40 mV)

2. Rapid Depolarization (Phase 0)

Threshold (-40 mV) reached

Opening of L-type Ca^{2+}
channels

Rapid influx of Ca^{2+} ions
(Upstroke to +5 mV)
(Note: Depolarization is
 Ca^{2+} -dependent, NOT Na^+ -
dependent)

3. Repolarization (Phase 3)

Peak reached (+5 mV)

Opening of K^+ channels &
prolonged K^+ efflux

Causes repolarization
down to maximum
diastolic potential (-60 mV)

Re-activates Na^+ leakage
→ Initiates next action
potential cycle